

Year 7 · Chapter 1 · 1.04 Adding integers: worked answers (teacher copy)

1.04 Adding integers (booklet p. 9)

Step 1: Adding a positive

E1 (WE DO) $-2 + 4$
 $= 2$

1 $-4 + 3$
 $= -1$

2 $-5 + 2$
 $= -3$

3 $-6 + 4$
 $= -2$

4 $-3 + 7$
 $= 4$

5 $-7 + 7$
 $= 0$

Step 2: Adding a negative

E2 (WE DO) $6 + (-2)$
 $= 6 - 2 \cdot = 4$

E3 (WE DO) $-2 + (-3)$
 $= -2 - 3 \cdot = -5$

6 $7 + (-3)$
 $= 7 - 3 \cdot = 4$

7 $5 + (-5)$
 $= 5 - 5 \cdot = 0$

8 $2 + (-6)$
 $= 2 - 6 \cdot = -4$

9 $4 + (-10)$
 $= 4 - 10 \cdot = -6$

10 $-1 + (-4)$
 $= -1 - 4 \cdot = -5$

11 $-3 + (-6)$
 $= -3 - 6 \cdot = -9$

Step 3: Zero pairs

E4 (WE DO) $-5 + 8$
 $8 - 5 = 3 \cdot + \text{has more: } 3$

E5 (WE DO) $6 + (-9)$
 $9 - 6 = 3 \cdot - \text{has more: } -3$

12 $-7 + 10$
 $10 - 7 = 3 \cdot + \text{has more: } 3$

13 $-9 + 4$
 $9 - 4 = 5 \cdot - \text{has more: } -5$

14 $8 + (-12)$
 $12 - 8 = 4 \cdot - \text{has more: } -4$

15 $-15 + 20$
 $20 - 15 = 5 \cdot + \text{has more: } 5$

16 $25 + (-40)$
 $40 - 25 = 15 \cdot - \text{has more: } -15$

17 $-36 + 50$
 $50 - 36 = 14 \cdot + \text{has more: } 14$

Step 4: Three numbers

E6 (WE DO) $1 + (-4) + 5$
 $= -3 + 5 \cdot = 2$

18 $2 + (-5) + 4$
 $= -3 + 4 \cdot = 1$

19 $-3 + 6 + (-2)$
 $= 3 + (-2) \cdot = 1$

20 $12 + (-15) + (-6)$
 $= -3 + (-6) \cdot = -9$

21 $-20 + 8 + 15$
 $= -12 + 15 \cdot = 3$

Step 5: Adding in real life

E7 (WE DO) It is -5°C . The temperature rises 8°C . What is the temperature now?
 $-5 + 8 \cdot = 3^{\circ}\text{C}$

22 It is -4°C at dawn. By lunch it has warmed up 9°C . What is the temperature at lunch?
 $-4 + 9 \cdot = 5^{\circ}\text{C}$

23 A lift is at level -2 . It goes up 5 floors. Which level is it on now?
 $-2 + 5 \cdot = \text{level } 3$

24 Sam has \$15 in the bank. Sam spends \$40 using credit. What is the balance now?
 $15 + (-40) \cdot = -\$25$

25 A diver is at -18 m . She swims up 7 m, then down 4 m. Where is she now?
 $-18 + 7 + (-4) \cdot = -11 + (-4) \cdot = -15\text{ m}$

Step 6: Is it right?

E8 (WE DO) Tom says $-6 + 2 = -8$. Is he right?
start at -6 , move 2 right $\cdot = -4 \cdot$ No. The answer is -4 .

26 Mia says $-8 + 5 = -13$. Is she right?
start at -8 , move 5 right $\cdot = -3 \cdot$ No. The answer is -3 .

27 Ben says $-7 + 3 + (-3) = -7$. Is he right?
 $3 + (-3)$ is a zero pair $\cdot -7 + 0 = -7 \cdot$ Yes. Ben is right.

28 Ali says $-4 + \square = 2$ means $\square = -2$. Is he right?
from -4 to 2 is 6 right $\cdot \square = 6 \cdot$ No. $\square$ is 6, not -2 .

29 Jo says adding always makes a number bigger. Is Jo right?
 $4 + (-6) = -2 \cdot -2$ is smaller than 4 $\cdot$ No. It can get smaller.